# Environment Setup for Satellite Server Installation

**Before installing Red Hat Satellite Server, ensure your system meets these prerequisites.**

## 1. System Details

- Operating System: RHEL 9
- Storage: 500 GB
- RAM: 34 GB
- CPU: 4 cores
- Hostname: Satellite-pd.example.com
- Subscription: Active Red Hat Satellite subscription
- Access: Root/admin
- DNS: Full forward and reverse resolution must work.
- Environment: Must install on a clean system with no other functions or conflicting users (like apache, postgres, pulp, redis, etc.).
- SELinux: Enabled (enforcing or permissive).
- Time Sync: System clock synchronized (e.g., using Chrony) to avoid SSL issues.
- FIPS Mode: Satellite supports DEFAULT and FIPS cryptographic policies, but FUTURE policy is not supported.

## 2. Storage Requirements

| Directory | Installation | Runtime |
|---|---|---|
| **/var/log** | **10 MB** | **10 GB** |
| **/var/lib/pgsql** | **100 MB** | **20 GB** |
| **/usr** | **10 GB** | **N/A** |
| **/opt/puppetlabs** | **500 MB** | **N/A** |
| **/var/lib/pulp** | **1 MB** | **300 GB** |

Guidelines:

- Mount /var on LVM for scalability.
- Use fast, low-latency storage for /var/lib/pulp and /var/lib/pgsql.
- /tmp must allow execution (exec) for Puppet.
- Avoid symbolic links for /var/lib/pulp.
- Manage logs using logrotate.
- For NFS mounts, set proper SELinux context to avoid blocks.

## 3. Supported Systems and Browsers

- Supported OS: Latest RHEL 9 (x86_64) only.
- Satellite must run on a dedicated system; no extra software.
- Supported browsers for UI: Firefox and Google Chrome.
- Supported languages: English, French, Japanese, Korean, Simplified Chinese.

## 4. Network and Firewall Requirements

Incoming Ports (Server):

- 53 TCP/UDP – DNS
- 443 TCP – HTTPS API and content sync
- 80 TCP – HTTP for registration
- 5910–5930 TCP – Browser console
- 8000 TCP – PXE/HTTP templates
- 9090 TCP – OpenSCAP, REX, Capsule API

Outgoing Ports (Server):

- 22 TCP – SSH for remote execution
- 53 TCP/UDP – DNS resolution
- 80/443 TCP – Content and repository sync

Use **firewall-cmd** to open required ports and make **persistent**.

# Setting up the environment

## Set the Hostname

We set a proper name for the server so it can be identified on the network.

```
[root@satellite-pd ~]# hostnamectl set-hostname satellite-
pd.example.com
```

**Verify the hostname:**

```
[root@satellite-pd ~]# hostnamectl
 Static hostname: satellite-pd.example.com
       Icon name: computer-vm
         Chassis: vm 🖴
      Machine ID: ********************************
         Boot ID: ********************************
  Virtualization: vmware
Operating System: Red Hat Enterprise Linux 9.6 (Plow)
     CPE OS Name: cpe:/o:redhat:enterprise_linux:9::baseos
          Kernel: Linux 5.14.0-427.13.1.el9_4.x86_64
    Architecture: x86-64
Firmware Version: VMW20
```

## Partition Setup (LVM)

Check Current Disks

- **lsblk** to check

```
[root@localhost ~]# lsblk
NAME          MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda             8:0    0   500G  0 disk
├─sda1          8:1    0   600M  0 part /boot/efi
├─sda2          8:2    0     1G  0 part /boot
└─sda3          8:3    0 498.4G  0 part
  ├─rhel-root 253:0    0    70G  0 lvm  /
  ├─rhel-swap 253:1    0  16.7G  0 lvm  [SWAP]
  └─rhel-home 253:2    0  50G  0 lvm  /home
```

```
sr0           11:0    1  1024M  0 rom
```

We created the following LVs:

- lv_log—Logs for Satellite
- lv_pupl—Content storage (Pulp)
- lv_pgsql—PostgreSQL database
- lv_tmp—Temporary files (/tmp)
- lv_puppet—Puppet data
- Home,usr and root already set with sufficient space and separate lv

**Commands Used**

- Creating lv

```
[root@localhost ~]# lvcreate -L 30G -n lv_pulp rhel
  Logical volume "lv_pulp" created.
```

- Creating filesystem

```
[root@localhost ~]# mkfs /dev/rhel/lv_pulp
mke2fs 1.46.5 (30-Dec-2021)
Discarding device blocks: done
Creating filesystem with 7864320 4k blocks and 1966080 inodes
Filesystem UUID: e873f35a-5044-41f7-b675-d148a9e9f218
Superblock backups stored on blocks:
      32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632,
2654208,
      4096000
Allocating group tables: done
Writing inode tables: ^C5/240
```

- Creating mount point

```
[root@localhost ~]# mkdir -p /var/lib/pulp
```

- Mounting permanent

```
[root@localhost ~]# vim /etc/fstab

UUID="************"     /var/lib/pulp   xfs     defaults        0       0
```

- Reload the daemon and mount -a

```
[root@localhost ~]# systemctl daemon-reload
[root@localhost ~]# mount -a
```

We repeated similar steps for lv_pulp, lv_log, lv_tmp, lv_puppet, and home with their respective directories.

**Final partition**

```
[root@satellite-pd ~]# lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda                   8:0    0   500G  0 disk
├─sda1                8:1    0   600M  0 part /boot/efi
├─sda2                8:2    0     1G  0 part /boot
└─sda3                8:3    0 498.4G  0 part
  ├─rhel-root       253:0    0    70G  0 lvm  /
  ├─rhel-swap       253:1    0  16.7G  0 lvm  [SWAP]
  ├─rhel-home       253:2    0    50G  0 lvm  /home
  ├─rhel-lv_pulp    253:3    0    30G  0 lvm  /var/lib/pulp
  ├─rhel-lv_pgsql   253:4    0    20G  0 lvm  /var/lib/pgsql
  ├─rhel-lv_log     253:5    0    10G  0 lvm  /var/log
  ├─rhel-lv_tmp     253:6    0    20G  0 lvm  /tmp
  └─rhel-lv_puppet  253:7    0     1G  0 lvm  /opt/puppetlabs
sr0                  11:0    1  1024M  0 rom
```

**lvs**

```
[root@satellite-pd ~]# lvs

  home      rhel -wi-ao----  50.00g
  lv_log    rhel -wi-ao----  10.00g
  lv_pgsql  rhel -wi-ao----  20.00g
  lv_pulp   rhel -wi-ao----  30.00g
  lv_puppet rhel -wi-ao----   1.00g
  lv_tmp    rhel -wi-ao----  20.00g
  root      rhel -wi-ao----  70.00g
```

```
  swap      rhel -wi-ao---- <16.70g
```

**DNS Server Setup**

Install DNS (BIND)

```
[root@satellite-pd ~]# yum install bind bind-utils -y
```

Enable and start service

```
[root@satellite-pd ~]# systemctl enable named.service
Created symlink /etc/systemd/system/multi-
user.target.wants/named.service →
/usr/lib/systemd/system/named.service.
[root@satellite-pd ~]# systemctl start named
[root@satellite-pd ~]# systemctl status named
```

Firewall rules setting

```
[root@satellite-pd ~]# firewall-cmd --add-service=dns --permanent
success
[root@satellite-pd ~]# firewall-cmd --add-port=53/tcp --permanent
success
[root@satellite-pd ~]# firewall-cmd --add-port=53/udp --permanent
success
```

Configure main dns file

```
[root@satellite-pd ~]# vim /etc/named.conf

 listen-on port 53 { 127.0.0.1;<ip>; };
 allow-query     { localhost; 192.168.0.0/16; any; };

//add zones
zone "example.com" IN {
    type master;
    file "forward.zone";
};
zone "96.168.192.in-addr.arpa" IN {
    type master;
    file "reverse.zone";
};
```

Create forward zone file

```
[root@satellite-pd ~]# vim /var/named/forward.zone

$TTL 1D
@   IN SOA satellite-pd.example.com. admin.example.com. (
        2025092901 ; serial
        1H         ; refresh
        10M        ; retry
        1W         ; expire
        1D )       ; minimum
    IN NS satellite-pd.example.com.
satellite-pd    IN A    192.168.96.220
```

Create reverse zone file

```
[root@satellite-pd ~]# vim /var/named/reverse.zone

$TTL 1D
@   IN SOA satellite-pd.example.com. admin.pd.example.com. (
        2025092902 ; serial  (increment serial)
        1H         ; refresh
        10M        ; retry
        1W         ; expire
        1D )       ; minimum
    IN NS satellite-pd.example.com.
; PTR record for reverse lookup
220 IN PTR satellite-pd.example.com.
```

Add Nameserver to **/etc/resolv.conf**

- **Note for persistent do it in nmtui in dns place add the ip of our dns server**

```
[root@satellite-pd ~]# vim /etc/resolv.conf

search example.com
nameserver 192.168.96.220 //add this
nameserver 8.8.8.8
```

Restart the service and test it

```
[root@satellite-pd ~]# systemctl restart named.service
[root@satellite-pd ~]# nslookup satellite-pd.example.com

Server:         192.168.96.220
Address:   192.168.96.220#53
```

```
Name: satellite-pd.example.com
Address: 192.168.96.220

[root@satellite-pd ~]# nslookup 192.168.96.220

220.96.168.192.in-addr.arpa name = satellite-pd.example.com.
```

## Sync time with the network (NTP)

Install Chrony

```
[root@satellite-pd ~]# dnf install chrony -y
```

Enable and Start the Service

```
[root@satellite-pd ~]# systemctl enable chronyd --now
[root@satellite-pd ~]# systemctl status chronyd
```

# Firewall and Services Configuration Documentation

This document explains the firewall configuration applied on the server, including incoming ports, outgoing ports, and services with commands and detailed explanations.

## 1. Incoming Ports Added

These are the ports opened for incoming traffic on the server:

| Port | Protocol | Purpose | Explanation |
|---|---|---|---|
| 448 | TCP | HTTPS for Pulp content delivery | Required for Satellite/Pulp to securely deliver content (RPM packages, metadata) to clients. |
| 80 | TCP | HTTP(repos, capsule/puppet communication) | Enables standard web access for repositories, Capsule servers, and Puppet agent communications over HTTP. |
| 53 | TCP/UDP | DNS (if server acts as DNS) | Used for resolving domain names. TCP is used for large queries like zone transfers, UDP for standard queries. |
| 8000 | TCP | HTTP(custom Satellite service / capsule) | Used by Red Hat Satellite for custom Capsule service communication. |
| 9090 | TCP | HTTPS(custom service / web UI) | Used for secure Satellite or application web interfaces (Cockpit or custom web UI). |

**Commands:**

Add incoming ports permanently

```
firewall-cmd --permanent --add-port=448/tcp
firewall-cmd --permanent --add-port=80/tcp
firewall-cmd --permanent --add-port=53/tcp
firewall-cmd --permanent --add-port=53/udp
firewall-cmd --permanent --add-port=8000/tcp
firewall-cmd --permanent --add-port=9090/tcp
```

**2. Outgoing Ports Added**

These are the ports allowed for outgoing connections from the server:

| Port | Protocol | Purpose | Explanation |
|---|---|---|---|
| 22 | TCP | SSH(remote host management) | Allows the server to initiate SSH connections to other servers for remote administration and automation. |
| 53 | TCP/UDP | DNS queries | Enables the server to query external DNS servers to resolve hostnames. |

| | | | |
|---|---|---|---|
| 443 | TCP | HTTPS (Red Hat CDN, external repositories) | Required for connecting to Red Hat CDN, fetching updates, and accessing secure external repositories. |

**Commands:**

Allow outgoing ports (example using rich rules if needed)

```
firewall-cmd --permanent --add-rich-rule='rule family="ipv4" port protocol="tcp" port="22" accept'
firewall-cmd --permanent --add-rich-rule='rule family="ipv4" port protocol="tcp" port="53" accept'
firewall-cmd --permanent --add-rich-rule='rule family="ipv4" port protocol="udp" port="53" accept'
firewall-cmd --permanent --add-rich-rule='rule family="ipv4" port protocol="tcp" port="443" accept'
```

## 3. Services Added

These are the services enabled in the firewall:

| Service | Ports Used | Purpose | Explanation |
|---|---|---|---|
| DNS | 53/TCP, 53/UDP | Name resolution | Required when the server itself acts as a DNS resolver or authoritative DNS server. |
| DHCP | 67/UDP, 68/UDP | Dynamic host configuration | Allows the server to provide/receive IP addresses dynamically (PXE boot environments). |
| TFTP | 69/UDP | File transfer for PXE boot | Used for network booting (PXE clients download boot images from server). |
| HTTP | 80/TCP | Web services & repository access | Provides unencrypted web access and repository services for clients. |
| HTTPS | 443/TCP | Secure web services | Ensures encrypted communication for web services, CDN access, and repositories. |

| | | | |
|---|---|---|---|
| Puppet Master | 8140/TCP | Puppet agent communication | Puppet agents use this port to connect to Puppet Master for configuration management. |

**Commands:**

```
# Add services permanently
firewall-cmd --permanent --add-service=dns
firewall-cmd --permanent --add-service=dhcp
firewall-cmd --permanent --add-service=tftp
firewall-cmd --permanent --add-service=http
firewall-cmd --permanent --add-service=https
```

Puppet is not a default firewalld service, so add by port

```
firewall-cmd --permanent --add-port=8140/tcp
```

### 4. Apply Changes

Once all ports and services are added, reload the firewall:

```
firewall-cmd --reload
```

To verify the applied rules and services:

```
firewall-cmd --list-all
```

### Summary

- **Incoming ports:** Opened to allow clients and services to communicate with this server.

- **Outgoing ports:** Enabled so the server can reach external repositories, DNS servers, and remote hosts.
- **Services:** Added to support network services (DNS, DHCP, TFTP, HTTP/HTTPS, Puppet).

This ensures the system can act as a Satellite/management server, provide DNS/DHCP/TFTP services, and securely fetch updates and content.

# Installing Satellite

Registering to Red Hat Subscription Management

```
[root@satellite-pd ~] subscription-manager register
Username: user_name
Password:
```

Configuring repositories

Disable all repositories:

```
[root@satellite-pd ~]# subscription-manager repos --disable "*"
```

Enable the following repositories:

```
[root@satellite-pd ~]# subscription-manager repos \
--enable=rhel-9-for-x86_64-baseos-rpms \
--enable=rhel-9-for-x86_64-appstream-rpms \
--enable=satellite-6.17-for-rhel-9-x86_64-rpms \
--enable=satellite-maintenance-6.17-for-rhel-9-x86_64-rpms
```

Verification

```
[root@satellite-pd ~]# dnf repolist enabled
```

Installing Satellite Server packages

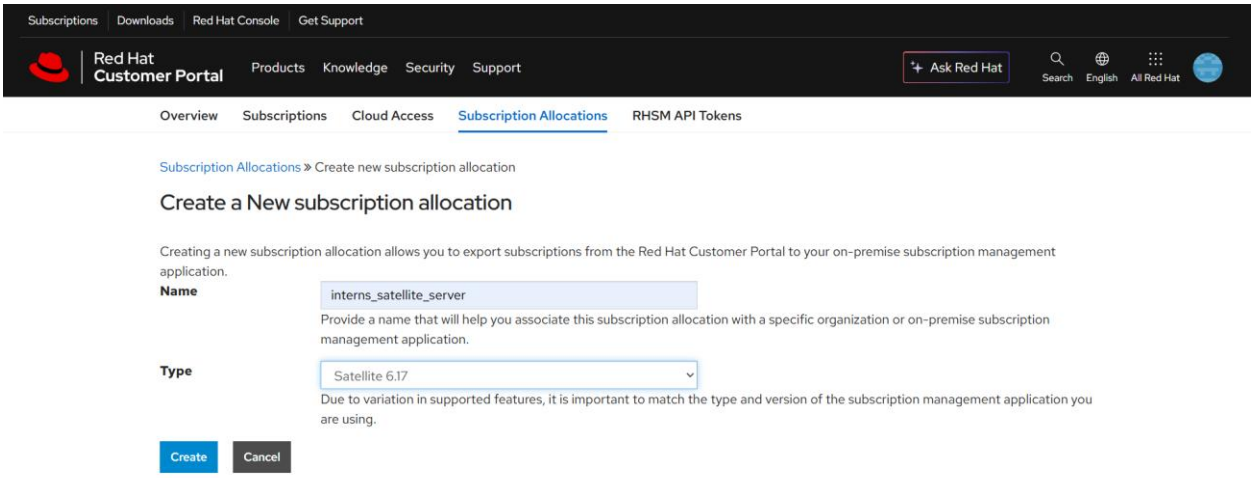
```
[root@satellite-pd ~]# dnf upgrade
[root@satellite-pd ~]# dnf install satellite
```

Configuring Satellite installation

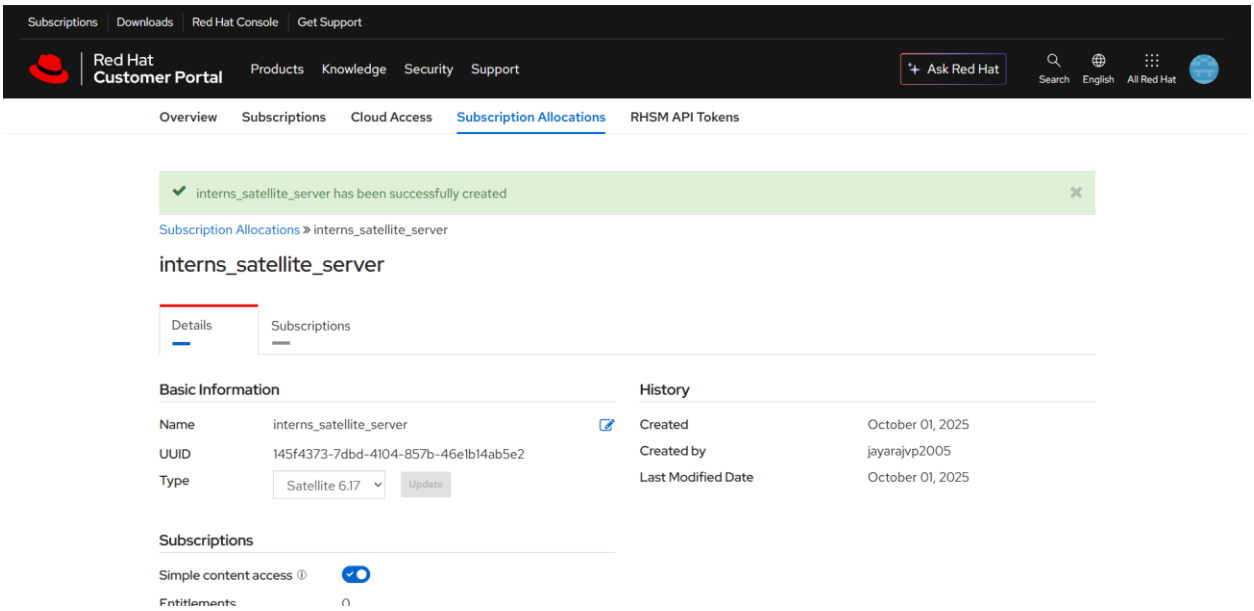
```
[root@satellite-pd ~]# satellite-installer --scenario satellite \
--foreman-initial-organization "My_Organization" \
--foreman-initial-location "My_Location" \
--foreman-initial-admin-username admin_user_name \
--foreman-initial-admin-password admin_password
```

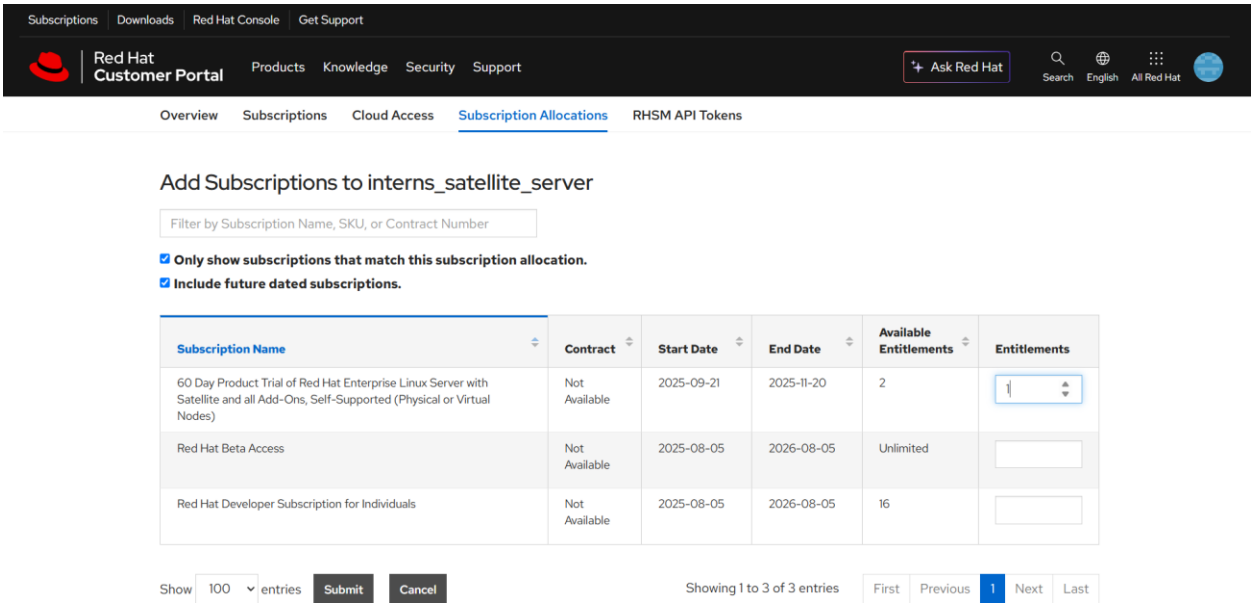
Importing a Red Hat subscription manifest into Satellite Server

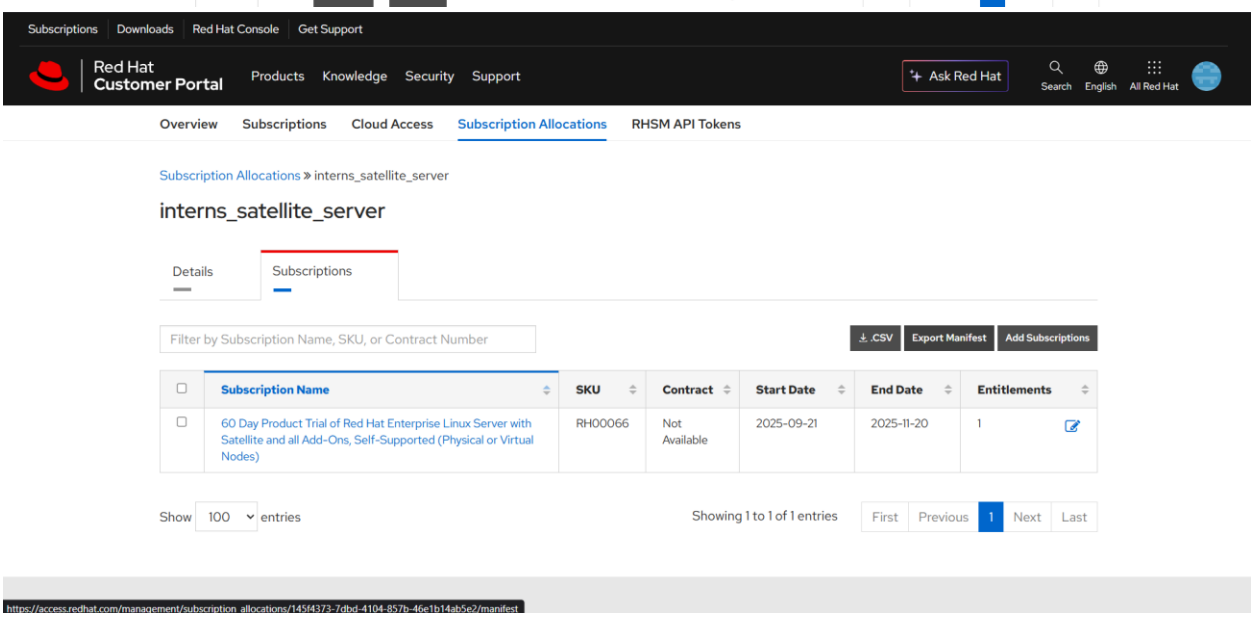
- Log in to Red Hat Customer Portal
- Navigate to Subscription Allocation
- Create a New Allocation
- Give in name, satellite version




- Select subscription and download the manifest file






Use pscp.exe to send the file to satellite machine

pscp.exe C:\path\to\manifest_file.zip root@<satellite-server-ip>:/root/

```
[root@satellite-pd ~]# hammer subscription upload \
--file ~/manifest_file.zip \
--organization "My_Organization"
```

```
[root@satellite-pd ~]# hammer subscription list --organization
"My_Organization"
```